

Assessment Name: Database system Assignment

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Introduction:

For this assignment, I designed and implemented a complete database system based on the scenario of the Oscar Film Awards from 2020–2025. I started by analysing the dataset and identifying suitable entities, attributes, and relationships to build an ER diagram using Chen notation. After finalising the design, I converted it into a relational schema and created all the required tables in MySQL with primary keys, foreign keys, and constraints.

Design of Database:

To design the database, I first studied the structure of the Oscar Award data then from the sample dataset, I identified the most important real-world objects that needed to be stored, such as ceremonies, locations, categories, films, nominations, and nominees; these became my main **entities**.

Why were these entities chosen:

Ceremony: Each Oscars event happens once per year and has attributes like year, ceremony number, and date.

Location: The Oscars are held at specific venues (e.g., Dolby Theatre, Union Station). Since a location can host multiple ceremonies, it made sense as a separate entity.

Category: Each award (e.g., Best Picture, Best Actor) is an important part of the dataset. I added a **composite attribute (CategoryLabel)** to show understanding of advanced ER features.

Film: Every nomination is tied to a film, so storing film title and release year was essential.

Nomination: This is the central fact of the database (a film being nominated for a category in a particular ceremony). It has a **composite primary key** (Year, CategoryName, Title).

Nominee: Represents the people nominated (actors, directors, etc.). This entity is a **weak entity** because it fully depends on the Nomination entity's PK.

Why these data types were selected

INT for years and ceremony numbers

VARCHAR for names, film titles, category labels

DATE for ceremony dates

TINYINT for IsWinner (boolean-like)

Derived attributes like FilmAge were kept as INT

All FK fields were assigned compatible data types to prevent integrity issues.

Why these constraints were used

Primary Keys ensure all main records remain unique.

Foreign Keys ensure referential integrity (no invalid nominations).

NOT NULL applied to essential fields.

Composite keys used to naturally represent multi-attribute uniqueness (e.g., nomination uniqueness).

Assumption Made:

- The Oscars occur once per year, and each year represents one ceremony.
- Film titles in the dataset are assumed to be unique.
- The dataset provided contains accurate mappings between films, nominees, categories, and years.
- A nomination cannot exist without a ceremony, a film, and a category.
- Ceremony dates were manually added from external sources (Wikipedia) because they were missing in the dataset.
- Some nominee roles were missing from the dataset, so the `role` attribute was stored but left NULL.

Entities and Attributes

Entity	Attributes
Ceremony	<u>Year</u> , CeremonyName, CeremonyDate
Location	<u>VenueName</u> , City, Country
Category	CategoryLabel (composite) categoryName canonicalName
Film	<u>Title</u> , ReleaseYear

Nomination	<u>Year, CategoryName, Title, IsWinner ,</u> FilmAge
Nominee	<u>PersonName, Role</u>

Relationships Sets And Cardinality

Relationship	Entity
Held_at	One ceremony happens at one location One location can host many ceremonies Ceremony to location → many-to-one relationship
Has	One ceremony has many nominations Each nomination belongs to exactly one ceremony Ceremony to nomination = one-to-many relationship
Used In	One CATEGORY appears in many nominations (every year, multiple films) One NOMINATION belongs to one CATEGORY Category to Nomination = one-to-many relationship
Gets	One FILM can have many nominations (multiple categories and/or years) Each NOMINATION is for one FILM Film to Nomination = one-to-many relationship

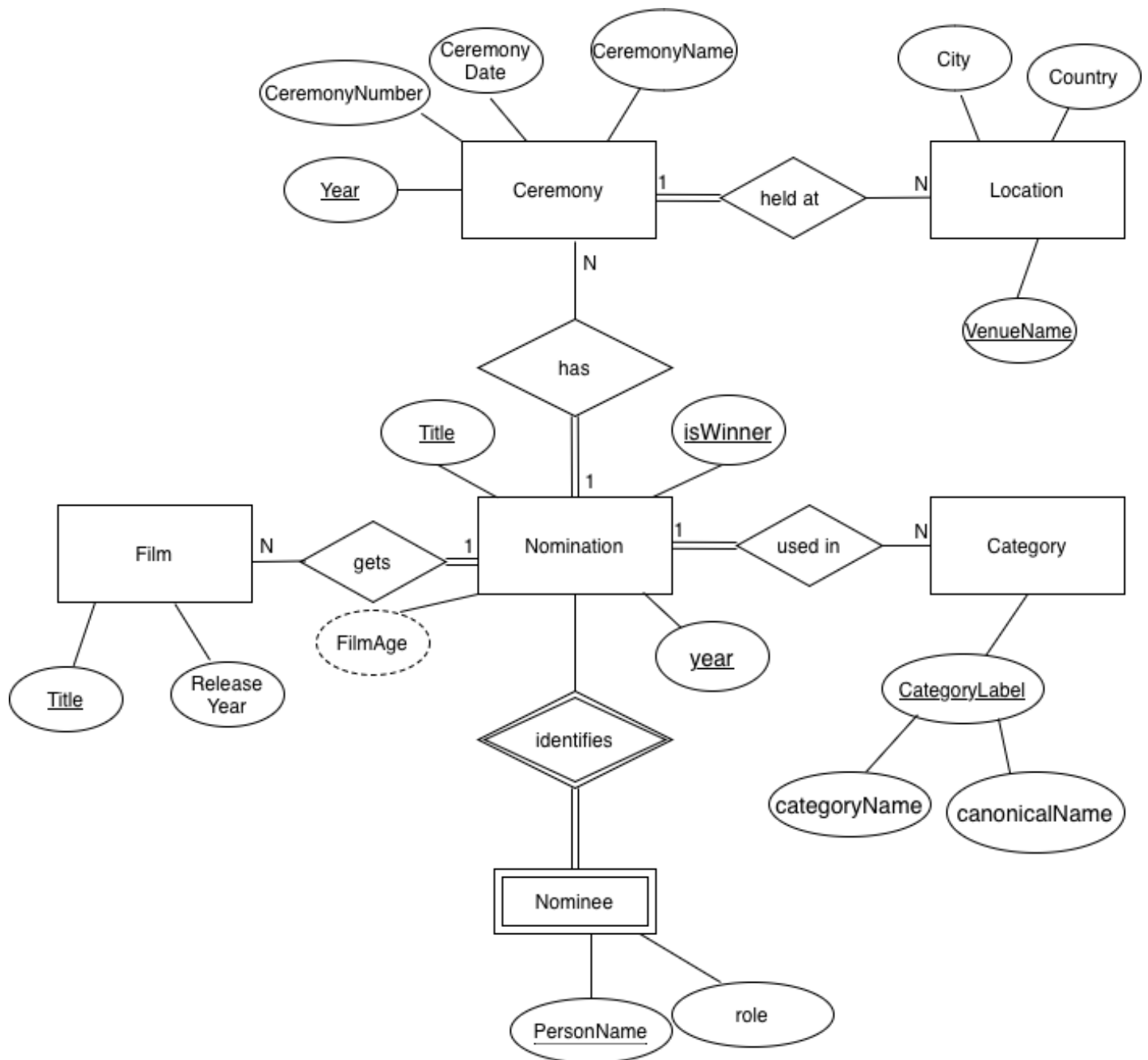
Identifies	One Nomination may have zero or many Nominees (e.g., Best Picture = none, Best Actor = one) Each Nominee depends on one Nomination Nomination to Nominee = One to Many Relationship
------------	--

Participation constraint

Relationship sets	Participation Constraint
Ceremony, Location	CEREMONY: total– every ceremony must have a location. LOCATION: partial– a location might not necessarily host any ceremony in your data.
Ceremony, Nomination	CEREMONY: partial (a ceremony can exist even before you add nominations) NOMINATION: total (a nomination must belong to a ceremony)
Category, Nomination	CATEGORY: partial A category can exist even if it has no nominations recorded for a particular year NOMINATION: total Every nomination must belong to one and only one category

Film, Nomination	FILM: partial A film can exist in the database even if it has not been nominated for any category. NOMINATION: total Every nomination must be associated with exactly one film, so a nomination cannot exist without a corresponding film.
Nomination, Nominee	NOMINATION: partial some nominations (e.g., Best Picture) have no person associated. NOMINEE: total a nominee cannot exist without a nomination.

ER diagram



1b)

CEREMONY(

Year PRIMARY KEY,

CeremonyNumber,

CeremonyName,

CeremonyDate,

LOCATION(

VenueName PRIMARY KEY,

City,

Country

)

CATEGORY(

categoryName PRIMARY KEY,

canonicalName

)

FILM(

Title PRIMARY KEY,

ReleaseYear

)

NOMINATION(

Year,

categoryName,

Title,

isWinner,

FilmAge, -- derived attribute

PRIMARY KEY (Year, categoryName, Title),

FOREIGN KEY (Year) REFERENCES CEREMONY(Year),

FOREIGN KEY (categoryName) REFERENCES CATEGORY(categoryName),

FOREIGN KEY (Title) REFERENCES FILM(Title)

NOMINEE(

 PersonName,

 role,

 Year,

 categoryName,

 Title,

 PRIMARY KEY (PersonName, Year, categoryName, Title),

 FOREIGN KEY (Year, categoryName, Title) REFERENCES NOMINATION(Year,
categoryName, Title)

1c) Data Description

Ceremony

Attribute	Data Type	Description	Constraints
Year	INT	Year of the Oscars ceremony	PRIMARY KEY, NOT NULL
CeremonyNumber	INT	Edition number of the ceremony	NOT NULL
CeremonyName	VARCHAR(100)	Official name of the ceremony	NOT NULL
CeremonyDate	DATE	Date the ceremony took place	NOT NULL

Location

Attribute	Data Type	Description	Constraints
VenueName	VARCHAR(100)	Unique venue identifier	PRIMARY KEY, NOT NULL
City	VARCHAR(60)	City of the venue	NOT NULL
Country	VARCHAR(60)	Country of the venue	NOT NULL

Category

Attribute	Data Type	Description	Constraints
categoryName	VARCHAR(120)	Name of award category	PRIMARY KEY, NOT NULL
canonicalName	VARCHAR(120)	Standardised form of category	NOT NULL

Film

Attribute	Data Type	Description	Constraints
Title	VARCHAR(150)	Title of the film	PRIMARY KEY, NOT NULL
ReleaseYear	INT	Year the film was released	NOT NULL

Nomination

Attribute	Data Type	Description	Constraints
Year	INT	Ceremony year	PK (composite), FK CEREMONY(Year), NOT NULL
CategoryName	VARCHAR(120)	Award category	PK (composite), FK CATEGORY(categoryName), NOT NULL
Title	VARCHAR(150)	Film title	PK (composite), FK → FILM(Title), NOT NULL
FilmAge	INT	Derived age of film at ceremony	Derived attribute (not stored)

Nominee (Weak Entity)

Attribute	Data Type	Description	Constraints
PersonName	VARCHAR(150)	Name of nominee (actor/director)	NOT NULL
Role	VARCHAR(60)	Role of nominee	NULL allowed
Year	INT	Ceremony year	PK (composite), FK → NOMINATION(Year), NOT NULL
categoryName	VARCHAR(120)	Award category	PK (composite), FK → NOMINATION(categoryName), NOT NULL
Title	VARCHAR(150)	Film title	PK (composite), FK → NOMINATION(Title), NOT NULL

Business Rules

1. Every ceremony must be held at exactly one location.
2. A film can exist without nominations; categories can exist without nominations.
3. A nomination **must** reference an existing ceremony, category, and film.
4. A nominee **cannot** exist without a nomination (weak entity constraint).
5. FilmAge is a derived attribute calculated as:

$$\text{FilmAge} = \text{CeremonyYear} - \text{ReleaseYear}$$
6. All primary keys must be unique and NOT NULL.
7. All foreign key values must match an existing referenced primary key (referential integrity).

2a) Implementation of Database

The database designed in Part 1 was implemented using MySQL Server.

A new database named **Nadim_22866899** was created as per instructions and all tables from the relational schema were implemented using CREATE TABLE statements with appropriate primary-key and foreign-key constraints. The implementation ensures referential integrity across all tables by enforcing foreign key relationships

```
mysql> Create database Nadim_22866899
-> ;
Query OK, 1 row affected (0.07 sec)

mysql> USE Nadim_22866899;
Database changed
```

```
Open  Implementation.sql  Ln 37, Col 33
~ /Desktop

1 /* MySQL file for creating tables */
2
3 -- create Location table
4 CREATE TABLE LOCATION (
5     VenueName VARCHAR(100) PRIMARY KEY,
6     City VARCHAR(60) NOT NULL,
7     Country VARCHAR(60) NOT NULL
8 );
9
10 -- create Ceremony table
11 CREATE TABLE CEREMONY (
12     Year INT PRIMARY KEY,
13     CeremonyNumber INT NOT NULL,SH
14     CeremonyName VARCHAR(100) NOT NULL,
15     CeremonyDate DATE NOT NULL,
16     VenueName VARCHAR(100) NOT NULL,
17     FOREIGN KEY (VenueName) REFERENCES LOCATION(VenueName)
18 );
19
20
21 -- create Category table
22 CREATE TABLE CATEGORY (
23     categoryName VARCHAR(120) PRIMARY KEY,
24     canonicalName VARCHAR(120) NOT NULL
25 );
26
27 -- create Film table
28 CREATE TABLE FILM (
29     Title VARCHAR(150) PRIMARY KEY,
30     ReleaseYear INT NOT NULL
31 );
```

```
Open  Implementation.sql  Ln 37, Col 33
~ /Desktop

32
33 -- Create Nomination Table
34 CREATE TABLE NOMINATION (
35     Year INT NOT NULL,
36     categoryName VARCHAR(120) NOT NULL,
37     Title VARCHAR(150) NOT NULL,
38     IsWinner BOOLEAN NOT NULL,
39     FilmAge INT, -- derived (may leave NULL, not
    physically stored)
40     PRIMARY KEY (Year, categoryName, Title),
41     FOREIGN KEY (Year) REFERENCES CEREMONY(Year),
42     FOREIGN KEY (categoryName) REFERENCES
    CATEGORY(categoryName),
43     FOREIGN KEY (Title) REFERENCES FILM(Title)
44 );
45
46 -- Create Nominee Table
47 CREATE TABLE NOMINEE (
48     PersonName VARCHAR(150) NOT NULL,
49     role VARCHAR(60),
50     Year INT NOT NULL,
51     categoryName VARCHAR(120) NOT NULL,
52     Title VARCHAR(150) NOT NULL,
53     PRIMARY KEY (PersonName, Year, categoryName, Title),
54     FOREIGN KEY (Year, categoryName, Title)
    REFERENCES NOMINATION(Year, categoryName, Title)
55 );
56
57
```

```
mysql> \. /home/nawal_nadim/Desktop/Implementation.sql
Query OK, 0 rows affected (0.05 sec)

Query OK, 0 rows affected (0.03 sec)

Query OK, 0 rows affected (0.02 sec)

Query OK, 0 rows affected (0.02 sec)

Query OK, 0 rows affected (0.03 sec)

Query OK, 0 rows affected (0.02 sec)
```

```
mysql> show tables;
```

```
+-----+
| Tables_in_Nadim_22866899 |
+-----+
| CATEGORY                  |
| CEREMONY                  |
| FILM                      |
| LOCATION                  |
| NOMINATION                |
| NOMINEE                   |
+-----+
6 rows in set (0.02 sec)
```

```
mysql> DESC CEREMONY; DESC CATEGORY;
```

```
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Year           | int           | NO   | PRI | NULL    |       |
| CeremonyNumber | int           | NO   |     | NULL    |       |
| CeremonyName   | varchar(100)  | NO   |     | NULL    |       |
| CeremonyDate   | date          | NO   |     | NULL    |       |
| VenueName      | varchar(100)  | NO   | MUL | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.02 sec)

+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| categoryName   | varchar(120)  | NO   | PRI | NULL    |       |
| canonicalName  | varchar(120)  | NO   |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> DESC FILM; DESC NOMINATION; DESC NOMINEE;
```

```
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Title          | varchar(150)  | NO   | PRI | NULL    |       |
| ReleaseYear    | int           | NO   |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.03 sec)

+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Year           | int           | NO   | PRI | NULL    |       |
| categoryName   | varchar(120)  | NO   | PRI | NULL    |       |
| Title          | varchar(150)  | NO   | PRI | NULL    |       |
| IsWinner       | tinyint(1)    | NO   |     | NULL    |       |
| FilmAge        | int           | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| PersonName     | varchar(150)  | NO   | PRI | NULL    |       |
| role           | varchar(60)   | YES  |     | NULL    |       |
| Year           | int           | NO   | PRI | NULL    |       |
| categoryName   | varchar(120)  | NO   | PRI | NULL    |       |
| Title          | varchar(150)  | NO   | PRI | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> DESC LOCATION;

+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| VenueName      | varchar(100)  | NO   | PRI | NULL    |       |
| City           | varchar(60)   | NO   |     | NULL    |       |
| Country        | varchar(60)   | NO   |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.03 sec)
```

2b) Loading Sample data

The sample data used in this project was derived from publicly available Academy Awards (Oscars) records. The original dataset was filtered to include only ceremonies from 2021 to 2024, and the filtered data was exported as a CSV file. This CSV was imported into MySQL through a staging table named *OSCAR_RAW* using the `LOAD DATA LOCAL INFILE` command. From this raw table, distinct values were inserted into the main relational tables (*CATEGORY*, *FILM*, *CEREMONY*, *LOCATION*, *NOMINATION*, and *NOMINEE*) using `INSERT INTO ... SELECT` queries. Missing attributes such as ceremony dates and venue names were added manually using verified public sources. Derived attributes (e.g., *FilmAge*) were computed using SQL update statements. All insertions were validated through select queries to ensure referential integrity.

```
mysql> LOAD DATA LOCAL INFILE '/home/nawal_nadim/Desktop/Oscars_Awards.csv'
-> INTO TABLE OSCAR_RAW
-> FIELDS TERMINATED BY ','
-> ENCLOSED BY '"'
-> LINES TERMINATED BY '\n'
-> IGNORE 1 LINES;
Query OK, 619 rows affected, 487 warnings (0.07 sec)
Records: 619 Deleted: 0 Skipped: 0 Warnings: 487
```

```
mysql> SELECT * FROM OSCAR_RAW LIMIT 10;
```

year_film	year_ceremony	ceremony	category	canon_category	name	film	winner
0	0	0	category	canon_category	name	film	winne
2020	2021	93	ACTOR IN A LEADING ROLE	ACTOR IN A LEADING ROLE	Riz Ahmed	Sound of Metal	FALSE
2020	2021	93	ACTOR IN A LEADING ROLE	ACTOR IN A LEADING ROLE	Chadwick Boseman	Ma Rainey's Black Bottom	FALSE
2020	2021	93	ACTOR IN A LEADING ROLE	ACTOR IN A LEADING ROLE	Anthony Hopkins	The Father	TRUE
2020	2021	93	ACTOR IN A LEADING ROLE	ACTOR IN A LEADING ROLE	Gary Oldman	Mank	FALSE
2020	2021	93	ACTOR IN A LEADING ROLE	ACTOR IN A LEADING ROLE	Steven Yeun	Minari	FALSE
2020	2021	93	ACTOR IN A SUPPORTING ROLE	ACTOR IN A SUPPORTING ROLE	Sacha Baron Cohen	The Trial of the Chicago 7	FALSE
2020	2021	93	ACTOR IN A SUPPORTING ROLE	ACTOR IN A SUPPORTING ROLE	Daniel Kaluuya	Judas and the Black Messiah	TRUE
2020	2021	93	ACTOR IN A SUPPORTING ROLE	ACTOR IN A SUPPORTING ROLE	Leslie Odom Jr.	One Night in Miami...	FALSE
2020	2021	93	ACTOR IN A SUPPORTING ROLE	ACTOR IN A SUPPORTING ROLE	Paul Raci	Sound of Metal	FALSE

```
10 rows in set (0.01 sec)
```

```
mysql> SELECT * FROM CEREMONY ORDER BY Year;
+-----+-----+-----+-----+-----+
| Year | CeremonyNumber | CeremonyName | CeremonyDate | VenueName |
+-----+-----+-----+-----+-----+
| 2021 | 93 | Academy Awards #93 | 2021-04-25 | Union Station |
| 2022 | 94 | Academy Awards #94 | 2022-03-27 | Dolby Theatre |
| 2023 | 95 | Academy Awards #95 | 2023-03-12 | Dolby Theatre |
| 2024 | 96 | Academy Awards #96 | 2024-03-10 | Dolby Theatre |
+-----+-----+-----+-----+-----+
4 rows in set (0.02 sec)
```

```
mysql> select * from LOCATION;
+-----+-----+-----+
| VenueName | City | Country |
+-----+-----+-----+
| Dolby Theatre | Los Angeles | USA |
| Union Station | Los Angeles | USA |
+-----+-----+-----+
2 rows in set (0.01 sec)
```

```
mysql> SELECT * FROM CATEGORY LIMIT 20;
+-----+-----+
| categoryName | canonicalName |
+-----+-----+
| ACTOR IN A LEADING ROLE | ACTOR IN A LEADING ROLE |
| ACTOR IN A SUPPORTING ROLE | ACTOR IN A SUPPORTING ROLE |
| ACTRESS IN A LEADING ROLE | ACTRESS IN A LEADING ROLE |
| ACTRESS IN A SUPPORTING ROLE | ACTRESS IN A SUPPORTING ROLE |
| ANIMATED FEATURE FILM | ANIMATED FEATURE FILM |
| AWARD OF COMMENDATION | AWARD OF COMMENDATION |
| BEST PICTURE | BEST PICTURE |
| CINEMATOGRAPHY | CINEMATOGRAPHY |
| COSTUME DESIGN | COSTUME DESIGN |
| DIRECTING | DIRECTING |
| DOCUMENTARY (Feature) | DOCUMENTARY (Feature) |
| DOCUMENTARY (Short Subject) | DOCUMENTARY (Short Subject) |
| DOCUMENTARY FEATURE FILM | DOCUMENTARY (Feature) |
| DOCUMENTARY SHORT FILM | DOCUMENTARY (Short Subject) |
| FILM EDITING | FILM EDITING |
| GORDON E. SAWYER AWARD | GORDON E. SAWYER AWARD |
| HONORARY AWARD | HONORARY AWARD |
| INTERNATIONAL FEATURE FILM | INTERNATIONAL FEATURE FILM |
| JEAN HERSHOLT HUMANITARIAN AWARD | JEAN HERSHOLT HUMANITARIAN AWARD |
| MAKEUP AND HAIRSTYLING | MAKEUP AND HAIRSTYLING |
+-----+-----+
20 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM FILM LIMIT 20;
+-----+-----+
| Title | ReleaseYear |
+-----+-----+
| 20 Days in Mariupol | 2023 |
| A Concerto Is a Conversation | 2020 |
| A House Made of Splinters | 2022 |
| A Love Song for Latasha | 2020 |
| A Shaun the Sheep Movie: Farmageddon | 2020 |
| Affairs of the Art | 2021 |
| Aftersun | 2022 |
| Ala Kachuu - Take and Run | 2021 |
| All Quiet on the Western Front | 2022 |
| All That Breathes | 2022 |
| All the Beauty and the Bloodshed | 2022 |
| American Fiction | 2023 |
| American Symphony | 2023 |
| An Irish Goodbye | 2022 |
| An Ostrich Told Me the World Is Fake and I Think I Believe It | 2022 |
| Anatomy of a Fall | 2023 |
| Another Round | 2020 |
| Argentina, 1985 | 2022 |
| Ascension | 2021 |
| Attica | 2021 |
+-----+-----+
20 rows in set (0.01 sec)
```



```
mysql> select * from NOMINEE LIMIT 20;
```

PersonName	role	Year	categoryName	Title
Aaron Sorkin	NULL	2021	WRITING (Original Screenplay)	The Trial of the Chicago 7
Adam McKay/David Sirota	NULL	2022	WRITING (Original Screenplay)	Don't Look Up
Adam McKay/Kevin Messick	NULL	2022	BEST PICTURE	Don't Look Up
Adam Stockhausen/Rena DeAngelo	NULL	2022	PRODUCTION DESIGN	West Side Story
Adrien Mørigeau/Amaury Ovisé	NULL	2021	SHORT FILM (Animated)	Genius Loci
Adrien Morot/Judy Chin/Annemarie Bradley	NULL	2023	MAKEUP AND HAIRSTYLING	The Whale
Ahmir "Questlove" Thompson/Joseph Patel/Robert Fyvolent/David Dinerstein	NULL	2022	DOCUMENTARY (Feature)	Summer of Soul (...Or, When the Revolution Could Not Be Televised)
Alan Baumgarten	NULL	2021	FILM EDITING	The Trial of the Chicago 7
Alberto Iglesias	NULL	2022	MUSIC (Original Score)	Parallel Mothers
Alberto Mielgo/Leo Sanchez	NULL	2022	SHORT FILM (Animated)	The Windshield Wiper
Alex Wuttke/Simone Coco/Jeff Sutherland/Neil Corbould	NULL	2024	VISUAL EFFECTS	Mission: Impossible - Dead Reckoning Part One
Alexander Nanau/Bianca Oana	NULL	2021	DOCUMENTARY (Feature)	Collective
Alexandra Byrne	NULL	2021	COSTUME DESIGN	Emma
Alice Rohrwacher/Alfonso Cuarón	NULL	2023	SHORT FILM (Live Action)	Le Pupille
Amanda Forbis/Wendy Tilby	NULL	2023	SHORT FILM (Animated)	The Flying Sailor
Amanda Seyfried	NULL	2021	ACTRESS IN A SUPPORTING ROLE	Mank
America Ferrera	NULL	2024	ACTRESS IN A SUPPORTING ROLE	Barbie
Ana de Armas	NULL	2023	ACTRESS IN A LEADING ROLE	Blonde
Ana López-Puigcerver/David Martel/Montse Ribó	NULL	2024	MAKEUP AND HAIRSTYLING	Society of the Snow
Anders Hammer/Charlotte Cook	NULL	2021	DOCUMENTARY (Short Subject)	Do Not Split

20 rows in set (0.01 sec)

```
mysql> select * from NOMINATION LIMIT 20;
```

Year	categoryName	Title	IsWinner	FilmAge
2021	ACTOR IN A LEADING ROLE	Ma Rainey's Black Bottom	0	NULL
2021	ACTOR IN A LEADING ROLE	Mank	0	NULL
2021	ACTOR IN A LEADING ROLE	Minari	0	NULL
2021	ACTOR IN A LEADING ROLE	Sound of Metal	0	NULL
2021	ACTOR IN A LEADING ROLE	The Father	1	NULL
2021	ACTOR IN A SUPPORTING ROLE	Judas and the Black Messiah	1	NULL
2021	ACTOR IN A SUPPORTING ROLE	One Night in Miami...	0	NULL
2021	ACTOR IN A SUPPORTING ROLE	Sound of Metal	0	NULL
2021	ACTOR IN A SUPPORTING ROLE	The Trial of the Chicago 7	0	NULL
2021	ACTRESS IN A LEADING ROLE	Ma Rainey's Black Bottom	0	NULL
2021	ACTRESS IN A LEADING ROLE	Nomadland	1	NULL
2021	ACTRESS IN A LEADING ROLE	Pieces of a Woman	0	NULL
2021	ACTRESS IN A LEADING ROLE	Promising Young Woman	0	NULL
2021	ACTRESS IN A LEADING ROLE	The United States vs. Billie Holiday	0	NULL
2021	ACTRESS IN A SUPPORTING ROLE	Borat Subsequent Moviefilm: Delivery of Prodigious Bribe to American Regime for Make Benefit Once Glorious Nation of Kazakhstan	0	NULL
2021	ACTRESS IN A SUPPORTING ROLE	Hillbilly Elegy	0	NULL
2021	ACTRESS IN A SUPPORTING ROLE	Mank	0	NULL
2021	ACTRESS IN A SUPPORTING ROLE	Minari	1	NULL
2021	ACTRESS IN A SUPPORTING ROLE	The Father	0	NULL
2021	ANIMATED FEATURE FILM	A Shaun the Sheep Movie: Farmageddon	0	NULL

20 rows in set (0.00 sec)

Questions , SQL queries and Output

Q1) List all the films released after the Year 2022

```
mysql> SELECT Title, ReleaseYear FROM FILM WHERE ReleaseYear > 2022;
```

Title	ReleaseYear
20 Days in Mariupol	2023
American Fiction	2023
American Symphony	2023
Anatomy of a Fall	2023
Barbie	2023
Bobbi Wine: The People's President	2023
El Conde	2023
Elemental	2023
Flamin' Hot	2023
Four Daughters	2023
Godzilla Minus One	2023
Golda	2023
Guardians of the Galaxy Vol. 3	2023
Indiana Jones and the Dial of Destiny	2023
Invincible	2023
Io Capitano	2023
Island in Between	2023
Killers of the Flower Moon	2023
Knight of Fortune	2023
Letter to a Pig	2023
Maestro	2023
May December	2023
Mission: Impossible - Dead Reckoning Part One	2023
Nai Nai & Wɔʔi Pɔʔ	2023
Napoleon	2023
Nimona	2023
Ninety-Five Senses	2023
Nyad	2023
Oppenheimer	2023
Our Uniform	2023
Pachyderme	2023
Past Lives	2023
Perfect Days	2023
Poor Things	2023
Red, White and Blue	2023
Robot Dreams	2023
Rustin	2023
Society of the Snow	2023
Spider-Man: Across the Spider-Verse	2023
The ABCs of Book Banning	2023
The After	2023
The Barber of Little Rock	2023
The Boy and the Heron	2023
The Color Purple	2023
The Creator	2023

Q2) List nominees whose name starts and ends with same letter

```
mysql> SELECT PersonName FROM NOMINEE WHERE LOWER(LEFT(PersonName,1)) = LOWER(RIGHT(PersonName,1));
```

PersonName
Alexander Nanau/Bianca Oana
Ren Klyce/Coya Elliott/David Parker
Nicole Kidman
Myron Kerstein/Andrew Weisblum
Nadia Stacey/Naomi Donne/Julia Vernon
Kelly Port/Chris Waegner/Scott Edelstein/Dan Sudick
Adam McKay/David Sirota
Todd Field/Alexandra Milchan/Scott Lambert
Dede Gardner/Jeremy Kleiner/Frances McDormand
Ruth Carter
Argentina
Ryan Tudhope/Seth Hill/Bryan Litson/Scott R. Fisher
Emma Stone
America Ferrera
Ed Guiney/Andrew Lowe/Yorgos Lanthimos/Emma Stone
Sean Wang/Sam Davis
Laurent Svonch
Luisa Abel
Nadia Stacey/Mark Coulier/Josh Weston
Stéphanie Clément/Marc Rius
Misan Harriman/Nicky Bentham

21 rows in set (0.01 sec)

Q3) List all ceremonies that took place in March.

```
mysql> SELECT Year, CeremonyDate, CeremonyName FROM CEREMONY WHERE MONTH(CeremonyDate) = 3;
```

Year	CeremonyDate	CeremonyName
2022	2022-03-27	Academy Awards #94
2023	2023-03-12	Academy Awards #95
2024	2024-03-10	Academy Awards #96

3 rows in set (0.00 sec)

Q4) Find the longest film title in the database.

```
mysql> SELECT Title FROM FILM ORDER BY LENGTH(Title) DESC LIMIT 1;
```

Title
Borat Subsequent Moviefilm: Delivery of Prodigious Bribe to American Regime for Make Benefit Once Glorious Nation of Kazakhstan

1 row in set (0.00 sec)

Q5) Find the weekday on which each ceremony took place.

```
mysql> SELECT Year, CeremonyDate, DAYNAME(CeremonyDate) AS Weekday FROM CEREMONY;
```

Year	CeremonyDate	Weekday
2021	2021-04-25	Sunday
2022	2022-03-27	Sunday
2023	2023-03-12	Sunday
2024	2024-03-10	Sunday

4 rows in set (0.01 sec)

Q6) Show each category and the number of films nominated in it.

```
mysql> SELECT categoryName, COUNT(*) AS TotalNominations
-> FROM NOMINATION
-> GROUP BY categoryName
-> ORDER BY TotalNominations DESC;
```

categoryName	TotalNominations
BEST PICTURE	38
WRITING (Adapted Screenplay)	20
SHORT FILM (Animated)	20
PRODUCTION DESIGN	20
MUSIC (Original Score)	20
MAKEUP AND HAIRSTYLING	20
INTERNATIONAL FEATURE FILM	20
FILM EDITING	20
SHORT FILM (Live Action)	20
SOUND	20
VISUAL EFFECTS	20
DIRECTING	20
COSTUME DESIGN	20
CINEMATOGRAPHY	20
WRITING (Original Screenplay)	20
ANIMATED FEATURE FILM	20
ACTRESS IN A LEADING ROLE	20
ACTOR IN A LEADING ROLE	20
ACTRESS IN A SUPPORTING ROLE	19
MUSIC (Original Song)	19
ACTOR IN A SUPPORTING ROLE	17
DOCUMENTARY SHORT FILM	10
DOCUMENTARY FEATURE FILM	10
DOCUMENTARY (Short Subject)	10
DOCUMENTARY (Feature)	10
AWARD OF COMMENDATION	1

26 rows in set (0.02 sec)

Q7) Find the film with the highest number of nominations.

```
mysql> SELECT Title, COUNT(*) AS NominationsCount
-> FROM NOMINATION
-> GROUP BY Title
-> ORDER BY NominationsCount DESC
-> LIMIT 1;
```

Title	NominationsCount
Oppenheimer	13

1 row in set (0.01 sec)

Q8) Show how many awards were won for each year

```
mysql> SELECT Year, SUM(IsWinner) AS TotalAwardsWon
-> FROM NOMINATION
-> GROUP BY Year
-> ORDER BY Year;
+-----+-----+
| Year | TotalAwardsWon |
+-----+-----+
| 2021 | 23 |
| 2022 | 23 |
| 2023 | 24 |
| 2024 | 23 |
+-----+-----+
4 rows in set (0.01 sec)
```

Q9) Find nominees who have won more than one Oscar award

```
mysql> SELECT      n.PersonName,      COUNT(*) AS TotalAwardsWon FROM NOMINEE n JOIN NOMINATION nom
ON n.Year
= nom.Year      AND n.categoryName = nom.categoryName      AND n.Title = nom.Title WHERE nom.IsWinner = 1
GROUP BY n.PersonName HAVING COUNT(*) > 1 ORDER BY TotalAwardsWon DESC;
+-----+-----+
| PersonName | TotalAwardsWon |
+-----+-----+
| Billie Eilish/Finneas O'Connell | 2 |
| Daniel Kwan/Daniel Scheinert | 2 |
+-----+-----+
2 rows in set (0.01 sec)
```

Q10) Show the top 5 categories with the highest number of Oscar wins.

```
mysql> SELECT categoryName, SUM(IsWinner) AS TotalWins FROM NOMINATION GROUP BY categoryName HAVING
SUM(IsWinner) > 0 ORDER BY TotalWins DESC LIMIT 5;
+-----+-----+
| categoryName | TotalWins |
+-----+-----+
| ACTOR IN A SUPPORTING ROLE | 4 |
| WRITING (Original Screenplay) | 4 |
| WRITING (Adapted Screenplay) | 4 |
| VISUAL EFFECTS | 4 |
| SOUND | 4 |
+-----+-----+
5 rows in set (0.02 sec)
```

Implementing Advanced Features

Stored Procedure 1:

This procedure inserts a new nomination into my NOMINATION table.

Before inserting, I used a simple SELECT INTO and an IF statement to check whether the same nomination already exists. If the record is new, it gets inserted.

This prevents accidental duplicate entries and makes the data cleaner.

The procedure uses IN parameters for year, category, title, and winner flag.

```
mysql> CREATE PROCEDURE AddNomination(
->     IN pYear INT,
->     IN pCategory VARCHAR(120),
->     IN pTitle VARCHAR(150),
->     IN pIsWinner TINYINT
-> )
-> BEGIN DECLARE existsCount INT DEFAULT 0;
-> SELECT COUNT(*) INTO existsCount
-> FROM NOMINATION WHERE Year = pYear AND categoryName = pCategory AND Title = pTitle; IF existsC
ount = 0 THEN
-> INSERT INTO NOMINATION (Year, categoryName, Title, IsWinner)
-> VALUES (pYear, pCategory, pTitle, pIsWinner);
-> END IF;
-> END $$
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL AddNomination(2023, 'DIRECTING', 'Barbie', 0);
Query OK, 1 row affected (0.00 sec)

mysql> SELECT * FROM NOMINATION WHERE Year = 2023 AND categoryName = 'DIRECTING' AND Title = 'Barbie';
+-----+-----+-----+-----+-----+
| Year | categoryName | Title | IsWinner | FilmAge |
+-----+-----+-----+-----+-----+
| 2023 | DIRECTING    | Barbie | 0        | NULL    |
+-----+-----+-----+-----+-----+
1 row in set (0.01 sec)

mysql> CALL AddNomination(2023, 'DIRECTING', 'Barbie', 0);
Query OK, 1 row affected (0.02 sec)

mysql> SELECT * FROM NOMINATION WHERE Year = 2023 AND categoryName = 'DIRECTING' AND Title = 'Barbie';
+-----+-----+-----+-----+-----+
| Year | categoryName | Title | IsWinner | FilmAge |
+-----+-----+-----+-----+-----+
| 2023 | DIRECTING    | Barbie | 0        | NULL    |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

Stored Procedure 2:

This procedure takes a year as input and returns all the winning nominees for that year.

It uses a JOIN between the NOMINEE and NOMINATION tables to match people with films and categories. I also applied a WHERE clause to filter only the winning entries, and an ORDER BY to show results neatly.

This procedure demonstrates using parameters and joins to produce meaningful reports from the database.

```
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE GetWinnersByYear(
->     IN pYear INT
-> )
-> BEGIN
->     SELECT
->         n.PersonName,
->         nom.Title,
->         nom.categoryName AS Category,
->         nom.Year
->     FROM NOMINEE n
->     JOIN NOMINATION nom
->         ON n.Year = nom.Year
->         AND n.categoryName = nom.categoryName
->         AND n.Title = nom.Title
->     WHERE nom.Year = pYear
->         AND nom.IsWinner = 1
->     ORDER BY nom.categoryName, n.PersonName;
-> END $$
```

Query OK, 0 rows affected (0.02 sec)

```
mysql>
mysql> DELIMITER ;
mysql> CALL GetWinnersByYear(2023);
```

PersonName	Title	Category	Year
Brendan Fraser	The Whale	ACTOR IN A LEADING ROLE	2023
Ke Huy Quan	Everything Everywhere All at Once	ACTOR IN A SUPPORTING ROLE	2023
Michelle Yeoh	Everything Everywhere All at Once	ACTRESS IN A LEADING ROLE	2023
Jamie Lee Curtis	Everything Everywhere All at Once	ACTRESS IN A SUPPORTING ROLE	2023
Stephanie Hsu	Everything Everywhere All at Once	ACTRESS IN A SUPPORTING ROLE	2023
Guillermo del Toro/Mark Gustafson/Gary Ungar/Alex Bulkley	Guillermo del Toro's Pinocchio	ANIMATED FEATURE FILM	2023
Ryan Laney	Welcome to Chechnya	AWARD OF COMMENDATION	2023
Daniel Kwan/Daniel Scheinert/Jonathan Wang	Everything Everywhere All at Once	BEST PICTURE	2023
James Friend	All Quiet on the Western Front	CINEMATOGRAPHY	2023
Ruth Carter	Black Panther: Wakanda Forever	COSTUME DESIGN	2023
Daniel Kwan/Daniel Scheinert	Everything Everywhere All at Once	DIRECTING	2023
Daniel Roher/Odessa Rae/Diane Becker/Melanie Miller/Shane Boris	Navalny	DOCUMENTARY FEATURE FILM	2023
Kartiki Gonsalves/Guneet Monga	The Elephant Whisperers	DOCUMENTARY SHORT FILM	2023
Paul Rogers	Everything Everywhere All at Once	FILM EDITING	2023
Germany	All Quiet on the Western Front	INTERNATIONAL FEATURE FILM	2023
Adrien Morot/Judy Chin/Annemarie Bradley	The Whale	MAKEUP AND HAIRSTYLING	2023
Volker Bertelmann	All Quiet on the Western Front	MUSIC (Original Score)	2023
M.M. Keeravaani/Chandrabose	RRR	MUSIC (Original Song)	2023
Christian M. Goldbeck/Ernestine Hipper	All Quiet on the Western Front	PRODUCTION DESIGN	2023
Charlie Mackesy/Matthew Freud	The Boy, the Mole, the Fox and the Horse	SHORT FILM (Animated)	2023
Tom Berkeley/Ross White	An Irish Goodbye	SHORT FILM (Live Action)	2023
Mark Weingarten/James H. Mather/Al Nelson/Chris Burdon/Mark Taylor	Top Gun: Maverick	SOUND	2023
Joe Letteri/Richard Banham/Eric Saindon/Daniel Barrett	Avatar: The Way of Water	VISUAL EFFECTS	2023
Sarah Polley	Women Talking	WRITING (Adapted Screenplay)	2023
Daniel Kwan/Daniel Scheinert	Everything Everywhere All at Once	WRITING (Original Screenplay)	2023

25 rows in set (0.03 sec)

Query OK, 0 rows affected (0.03 sec)

Trigger 1:

I added a trigger that automatically calculates the FilmAge attribute when a new NOMINATION row is inserted. The trigger looks up the film's ReleaseYear from the FILM table, subtracts it from the nomination Year and stores the computed value into NEW.FilmAge

It shows the use of derived attributes and ensures FilmAge is always consistent and accurate without manual calculation.

```
mysql> DELIMITER $$
mysql>
mysql> CREATE TRIGGER before_nomination_insert
-> BEFORE INSERT ON NOMINATION
-> FOR EACH ROW
-> BEGIN
->     DECLARE filmYear INT;
->
->     -- Get film's release year
->     SELECT ReleaseYear
->     INTO filmYear
->     FROM FILM
->     WHERE Title = NEW.Title;
->
->     -- Calculate derived attribute FilmAge
->     SET NEW.FilmAge = NEW.Year - filmYear;
-> END$$
```

Query OK, 0 rows affected (0.07 sec)

```
mysql> INSERT INTO NOMINATION (Year, categoryName, Title, IsWinner) VALUES (2024, 'BEST PICTURE',
'Dune', 0);
```

Query OK, 1 row affected (0.02 sec)

```
mysql> SELECT Year, categoryName, Title, FilmAge FROM NOMINATION WHERE Title='Dune' AND Year=2024
AND categoryName='BEST PICTURE';
```

```
+-----+-----+-----+-----+
| Year | categoryName | Title | FilmAge |
+-----+-----+-----+-----+
| 2024 | BEST PICTURE | Dune  |        3 |
+-----+-----+-----+-----+
```

1 row in set (0.01 sec)

View_Winners is a simple view I created to make it easier to retrieve all winning nominations from my database. Instead of writing the full SELECT query each time, I can query the view directly. This helps with readability and keeps queries cleaner. The view filters the NOMINATION table and returns only the rows where IsWinner = 1.

Index: I created an index on the **categoryName** column in the NOMINATION table.

Since many of my queries filter nominations by category, this index helps MySQL find matching rows faster. This improves performance and shows the use of indexing in a real scenario.

```
mysql> CREATE INDEX idx_categoryName  
-> ON NOMINATION (categoryName);  
Query OK, 0 rows affected (0.10 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> SHOW INDEX FROM NOMINATION;
```

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment	Visible	Expression
NOMINATION	0	PRIMARY	1	Year	A	4	NULL	NULL		BTREE			YES	NULL
NOMINATION	0	PRIMARY	2	categoryName	A	93	NULL	NULL		BTREE			YES	NULL
NOMINATION	0	PRIMARY	3	Title	A	474	NULL	NULL		BTREE			YES	NULL
NOMINATION	1	Title	1	Title	A	217	NULL	NULL		BTREE			YES	NULL
NOMINATION	1	idx_categoryName	1	categoryName	A	26	NULL	NULL		BTREE			YES	NULL

```
5 rows in set (0.03 sec)
```

```
mysql> EXPLAIN SELECT *  
-> FROM NOMINATION  
-> WHERE categoryName = 'BEST PICTURE';
```

id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	NOMINATION	NULL	ref	idx_categoryName	idx_categoryName	482	const	39	100.00	NULL

```
1 row in set, 1 warning (0.02 sec)
```


5) I connected my MySQL database to Python using the mysql.connector library. After creating the connection, I ran some of the SQL queries from Part 3 inside Python to show that I can retrieve data programmatically. First, I displayed a sample set of results (e.g., the first 5 films and all nominations for “Oppenheimer”). Then, I demonstrated data manipulation by inserting a new film and a new nomination through Python. After inserting, I ran another SELECT query to confirm that the new record was successfully added to the database.

```
GNU nano 7.2                                con1.py *
import mysql.connector

try:
    connection = mysql.connector.connect(
        host='localhost',
        user='root',
        password='nawal123',
        database='Nadim_22866899',
        ssl_disabled=True
    )

    if connection.is_connected():
        print("Connected Successfully\n")

    cursor = connection.cursor(dictionary=True)

    q1 = "SELECT Title, ReleaseYear FROM FILM LIMIT 5"
    cursor.execute(q1)
    result = cursor.fetchall()

    print("First 5 films:")
    for row in result:
        print(row)

    q2 = "SELECT * FROM NOMINATION WHERE Title = %(t)s"
    cursor.execute(q2, {"t": "Oppenheimer"})
    rows = cursor.fetchall()

    print("\nNominations for Oppenheimer:")
    for r in rows:
        print(r)

    print("\nInserting a test film + nomination...")

    cursor.execute(
        "INSERT INTO FILM (Title, ReleaseYear) VALUES (%s, %s)",
        ("Python Demo Film", 2024)
    )

    cursor.execute(
        """INSERT INTO NOMINATION (Year, categoryName, Title, IsWinner)
        VALUES (%s, %s, %s, %s)""",
        (2024, "BEST PICTURE", "Python Demo Film", 0)
    )

    connection.commit()
```

```
nawal_nadim@nawal-nadim:~$ nano con1.py
nawal_nadim@nawal-nadim:~$ python3 con1.py
Connected Successfully

First 5 films:
{'Title': '20 Days in Mariupol', 'ReleaseYear': 2023}
{'Title': 'A Concerto Is a Conversation', 'ReleaseYear': 2020}
{'Title': 'A House Made of Splinters', 'ReleaseYear': 2022}
{'Title': 'A Love Song for Latasha', 'ReleaseYear': 2020}
{'Title': 'A Shaun the Sheep Movie: Farmageddon', 'ReleaseYear': 2020}

Nominations for Oppenheimer:
{'Year': 2024, 'categoryName': 'ACTOR IN A LEADING ROLE', 'Title': 'Oppenheimer', 'IsWinner': 1, 'FilmAge': None}
{'Year': 2024, 'categoryName': 'ACTOR IN A SUPPORTING ROLE', 'Title': 'Oppenheimer', 'IsWinner': 1, 'FilmAge': None}
{'Year': 2024, 'categoryName': 'ACTRESS IN A SUPPORTING ROLE', 'Title': 'Oppenheimer', 'IsWinner': 0, 'FilmAge': None}
{'Year': 2024, 'categoryName': 'BEST PICTURE', 'Title': 'Oppenheimer', 'IsWinner': 1, 'FilmAge': None}
{'Year': 2024, 'categoryName': 'CINEMATOGRAPHY', 'Title': 'Oppenheimer', 'IsWinner': 1, 'FilmAge': None}
{'Year': 2024, 'categoryName': 'COSTUME DESIGN', 'Title': 'Oppenheimer', 'IsWinner': 0, 'FilmAge': None}
{'Year': 2024, 'categoryName': 'DIRECTING', 'Title': 'Oppenheimer', 'IsWinner': 1, 'FilmAge': None}
{'Year': 2024, 'categoryName': 'FILM EDITING', 'Title': 'Oppenheimer', 'IsWinner': 1, 'FilmAge': None}
{'Year': 2024, 'categoryName': 'MAKEUP AND HAIRSTYLING', 'Title': 'Oppenheimer', 'IsWinner': 0, 'FilmAge': None}
{'Year': 2024, 'categoryName': 'MUSIC (Original Score)', 'Title': 'Oppenheimer', 'IsWinner': 1, 'FilmAge': None}
{'Year': 2024, 'categoryName': 'PRODUCTION DESIGN', 'Title': 'Oppenheimer', 'IsWinner': 0, 'FilmAge': None}
{'Year': 2024, 'categoryName': 'SOUND', 'Title': 'Oppenheimer', 'IsWinner': 0, 'FilmAge': None}
{'Year': 2024, 'categoryName': 'WRITING (Adapted Screenplay)', 'Title': 'Oppenheimer', 'IsWinner': 0, 'FilmAge': None}

Inserting a test film + nomination...
Insert successful
[{'Year': 2024, 'categoryName': 'BEST PICTURE', 'Title': 'Python Demo Film', 'IsWinner': 0, 'FilmAge': 0}]

Connection Closed
```

Discussion:

In this assignment, I was able to design a full database system from scratch, implement it in MySQL, insert and manage data, create advanced features like triggers, views, and an index, and finally connect the database to Python to run queries and perform insert operations.

Overall, the project helped me understand the complete process of database development from design all the way to application-level interaction.

The biggest challenge I faced was at the start; since we had to create the ER diagram ourselves, it took me a long time to identify the correct entities, attributes, and relationships from the scenario and then sorting the raw information into a proper structure was harder than expected, and I had to revise the design several times before everything fit logically.

Another challenge was inserting the sample data into SQL because of foreign key constraints, I couldn't just insert information randomly. I faced several errors before figuring out the correct workflow for entering data smoothly then setting primary keys and foreign keys correctly also took careful planning. I also had small limitations while working on triggers and testing calculated fields, where I had to check my logic multiple times to ensure outputs were correct. If I were to improve this project, I would try adding more validation rules, more advanced triggers.

Overall, this assignment helped me strengthen my understanding of database design, SQL implementation, and real-world integration using Python. Despite the challenges, completing each part gave me a much clearer idea of how full database systems are built and used.